

Cloud Platforms for AI

*How AWS, Google Cloud, and Azure power the tools that build, train, and run
AI systems*

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What Is Cloud Computing for AI?

- ▶ Cloud computing means renting computing power, storage, and software over the internet instead of buying physical hardware.
- ▶ For AI, that matters a lot — training a machine learning model can require thousands of hours of processing on specialized chips called GPUs.
- ▶ Without the cloud, only large companies could afford AI. Cloud platforms make it accessible to startups, researchers, and students.
- ▶ The three biggest cloud providers for AI are:
 - ▶ Amazon Web Services (AWS)
 - ▶ Google Cloud Platform (GCP)
 - ▶ Microsoft Azure
- ▶ Each offers tools to collect data, train models, deploy them, and monitor them in production.

Why Use the Cloud for AI?

Scale On Demand

Need 100 GPUs for a week? Rent them. Done with training? Stop paying. No hardware investment needed.

Speed

Cloud GPUs and TPUs can cut training time from weeks down to hours. Google's TPUs were designed specifically for AI.

Pre-Built AI Services

All three clouds offer ready-to-use AI: image recognition, speech-to-text, translation, fraud detection, and more.

Global Reach

Deploy your AI model in data centers around the world so users everywhere get fast responses.

Managed MLOps

Tools like AWS SageMaker and Azure ML handle the pipeline: data prep, training, testing, deployment, monitoring.

Cost Control

Pay only for what you use. Set budgets, use spot instances (cheaper idle servers), and auto-scale up or down.

The Three Major Cloud Platforms

AWS

Amazon Web Services

- Largest cloud provider worldwide
- SageMaker: full ML pipeline
- Rekognition (image AI)
- Comprehend (text/NLP)
- Bedrock: Claude, Llama, Titan models
- Strong enterprise adoption

Google Cloud

GCP

- Invented the Transformer architecture
- TPUs: chips built specifically for AI
- Vertex AI: unified ML platform
- BigQuery: AI on very large datasets
- Gemini models integrated throughout
- Best for research and NLP work

Azure

Microsoft Azure

- Deep OpenAI partnership
- Azure OpenAI: GPT-4, DALL-E
- Azure ML: end-to-end pipeline
- Cognitive Services: vision, speech
- Best for Microsoft 365 environments
- Strong in enterprise and healthcare

Platform Comparison at a Glance

Category	AWS	Google Cloud	Azure
ML Platform	SageMaker	Vertex AI	Azure ML
AI Hardware	NVIDIA GPUs	TPUs + GPUs	NVIDIA GPUs
Pre-built AI APIs	Rekognition, Comprehend, Polly	Vision AI, Natural Language, Speech	Cognitive Services
Foundation Models	Bedrock (Claude, Llama, Titan)	Gemini	OpenAI (GPT-4, DALL-E)
Best Known For	Breadth, market share	Research, NLP, TPUs	Enterprise, Office 365
Pricing Model	Pay-as-you-go, spot instances	Sustained use discounts	Hybrid + reserved pricing

MLOps: Taking a Model from Training to Production

MLOps stands for Machine Learning Operations. It is the process of building, testing, deploying, and monitoring AI models so they work reliably in the real world, not just in a notebook.

1. Data Ingestion

Raw data is collected and stored in cloud storage buckets (S3, GCS, Azure Blob).

2. Data Prep

Data is cleaned, labeled, and transformed. Tools: AWS Glue, Dataflow, Azure Data Factory.

3. Model Training

The algorithm learns patterns from data on GPU/TPU clusters. Can take hours or days.

4. Model Evaluation

Accuracy, precision, recall, and F1 score are measured. Bad models do not move forward.

5. Deployment

The model is packaged and served as an API endpoint. Users send data in, predictions come out.

6. Monitoring

Real-world performance is tracked. If the model degrades over time (data drift), it gets retrained.

How Cloud AI Pricing Works

- ▶ All three clouds use pay-as-you-go pricing: you pay only for the compute, storage, and API calls you actually use.
- ▶ Key cost factors for AI workloads:
 - ▶ Compute: GPU/TPU hours are the biggest expense in training. A high-end GPU can cost \$3 to \$30+ per hour.
 - ▶ Storage: Storing large datasets (images, video, sensor data) is charged per GB/month.
 - ▶ API calls: Pre-built AI services charge per request (e.g. \$0.001 per image analyzed).
 - ▶ Egress: Moving data out of the cloud costs money. Moving it in is usually free.
- ▶ Ways to cut costs:
 - ▶ Spot / Preemptible instances: Unused cloud capacity sold cheap (60-90% discount). Can be interrupted.
 - ▶ Reserved instances: Commit to 1-3 years and get up to 72% off on-demand prices.
 - ▶ Sustained use discounts: Google Cloud automatically discounts the more you use in a month.
 - ▶ Auto-scaling: Scale down to zero when not in use. Never pay for idle servers.

Case Study 1 of 2 | Netflix + Amazon Web Services

What Netflix Did

- ▶ Netflix moved its entire infrastructure to AWS in 2016, closing its last data center.
- ▶ It uses AWS SageMaker to train and deploy its recommendation engine, which suggests what you should watch next.
- ▶ The model processes viewing history, search patterns, ratings, time of day, and even what device you are using.
- ▶ AWS EC2 (compute) and S3 (storage) handle billions of streaming events per day.
- ▶ AWS Lambda (serverless) runs lightweight AI tasks like thumbnail personalization without needing a dedicated server.

80%

of content watched
comes from AI
recommendations

\$1B+

saved per year by
reducing churn through
better suggestions

247M+

subscribers served
across 190 countries
from AWS infrastructure

What Mayo Clinic Did

- ▶ Mayo Clinic partnered with Microsoft to build an AI clinical platform on Azure.
- ▶ They use Azure Machine Learning to train models that analyze patient records, lab results, imaging scans, and clinical notes.
- ▶ Azure OpenAI Service helps generate structured summaries from unstructured doctor notes, saving physicians time.
- ▶ Azure's HIPAA-compliant infrastructure keeps patient data protected under healthcare privacy law.
- ▶ The platform also accelerated COVID-19 research by finding patterns in patient outcomes across millions of records.

72%

faster review of clinical
trial eligibility using AI
screening

Millions

of patient records
analyzed to improve
diagnosis accuracy

HIPAA

compliant cloud setup
trusted for sensitive
healthcare data

Challenges and Risks of Cloud AI

Cost Can Explode

Leaving GPU instances running when not needed can result in massive surprise bills. Cloud cost management is a real discipline.

Vendor Lock-In

Once you build deeply on one platform's tools (SageMaker, Vertex AI), switching to another is expensive and time-consuming.

Data Privacy

Sending sensitive data to a third-party cloud raises legal and ethical questions, especially in healthcare, finance, and government.

Latency

Cloud AI adds a network round-trip. For real-time applications like self-driving cars or robotics, edge computing may be needed.

Skills Gap

Cloud AI platforms are powerful but complex. Teams need training in MLOps, cloud architecture, and DevOps, not just data science.

Model Reliability

AI models can degrade over time as real-world data changes. Without proper monitoring, you may not notice until something goes wrong.

Key Takeaways

- ▶ Cloud platforms (AWS, Google Cloud, Azure) remove the barrier to building AI by providing scalable compute, storage, and pre-built models on demand.
- ▶ Each platform has a specialty: AWS for breadth and market reach, Google Cloud for research and TPUs, Azure for enterprise and OpenAI integration.
- ▶ MLOps pipelines take an AI model from a notebook all the way to a live product that real users interact with.
- ▶ Pricing is flexible but requires discipline: spot instances, auto-scaling, and budget alerts are essential to avoid surprise costs.
- ▶ Real-world results are significant: Netflix saved over \$1 billion in customer retention; Mayo Clinic cut clinical trial screening time by 72%.
- ▶ Challenges are real: cost overruns, vendor lock-in, data privacy, and the need for ongoing monitoring all require careful planning.

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